

PUBLISHED

Question	Answer	Marks	Guidance
2(a)	$p = \frac{e^{-2.8}(2.8)^3}{3!} \times 100 = 22.248$	B1	Or by subtraction. Allow 22.2 or 22.3.
	$q = \frac{(29 - 22.25)^2}{22.25} = 2.05$	B1	AWRT 2.05.
		2	
2(b)	H_0: Po(2.8) fits the data. H_1: Po(2.8) does not fit the data. OR H_0: Po(2.8) is a satisfactory model for the number of cars (entering the car park). H_1: Po(2.8) is not a satisfactory model for the number of cars (entering the car park).	B1	Must mention distribution and data/number of cars.
	$2.739 + 0.241 + 2.152 + '2.049' + 0.425 + 3.753 + 0.037 \quad [= 11.396]$	M1	Sum of chi-square contributions using <i>their</i> q . May be implied by 11.4.
	'11.4' < 12.59, accept H_0 / do not reject H_0 / not significant.	M1	Compare <i>their</i> 11.4 with 12.59 and appropriate result (may be in terms of H_1). Test result may be implied by an attempt at an appropriate conclusion in context ignoring hypotheses.
	Insufficient evidence to suggest that Po(2.8) is not a good fit to the data. OR Insufficient evidence to suggest that the manager's claim is false.	A1	Correct conclusion from correct working ignoring hypotheses. In context. Level of uncertainty in language is used (for example, not 'prove'). Allow 'Not enough evidence that / to show / conclude that...' Do not accept statements such as 'there is sufficient evidence to suggest...' or 'no evidence.....'
		4	